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Exosomal pHERV-W ENV as a dynamic biomarker for relapse prediction and prognosis in multiple sclerosis

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Abstract

Background Emerging evidence implicates the envelope protein of human endogenous retrovirus type W (pHERV-W ENV) in the pathogenesis of multiple sclerosis (MS); however, its clinical utility as a biomarker remains incompletely defined. In this study, we demonstrate that pHERV-W ENV is enriched on serum-derived exosomes in patients with relapsing-remitting MS (RRMS), where it reflects both current disease activity and future risk.

Methods Exosomal pHERV-W ENV levels were profiled across three cohorts: (1) a cross-sectional cohort including healthy controls, stable MS patients, and patients with active disease; (2) a longitudinal RRMS cohort with samples collected before, during, and after episodes of disease activity; and (3) a retrospective longitudinal cohort of newly diagnosed RRMS patients with three years of follow-up.

Results We observed an increase in exosomal pHERV-W ENV expression during active disease, followed by a decline at follow-up. Levels of pHERV-W ENV expression during active disease positively correlated with Epstein-Barr virus (EBV)-derived latent membrane protein 2 A (LMP2A) expression, supporting an association between endogenous retroviral expression and EBV activity. Flow cytometric profiling of immune cell subsets identified monocytes and B cells as the predominant sources of circulating pHERV-W ENV. Longitudinal profiling further revealed coordinated immune and vascular activation during active disease, including a significant positive correlation between exosomal pHERV-W ENV and the costimulatory molecule CD40. Notably, high baseline levels of exosomal pHERV-W ENV were associated with greater total disease activity events, defined as the sum of clinical relapses and MRI activity (new/enlarging T2 or gadolinium-enhancing (Gd+) lesions) over three years, as well as with accelerated loss of “No Evidence of Disease Activity” (NEDA) status over time.

Conclusion These findings suggest that exosomal pHERV-W ENV expression may serve as a dynamic, non-invasive biomarker of disease activity with prognostic value in MS.

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Introduction

Multiple sclerosis (MS) is a chronic demyelinating disorder of the central nervous system (CNS) and a leading cause of neurological disability in young adults [1]. While genetic susceptibility accounts for ~ 30% of disease risk, environmental factors particularly Epstein-Barr virus (EBV) infection are strongly implicated as triggers [2, 3]. Concurrently, dysregulated expression of human endogenous retrovirus type W (HERV-W) is recognized as a significant contributor to MS pathogenesis and progression [4–6]. HERV-W envelope protein (pHERV-W ENV) exhibits direct neuropathogenic effects: it promotes immune-mediated inflammation [7], inhibits remyelination in MS [8], and exacerbates disease in experimental autoimmune encephalomyelitis models [9]. Critically, pHERV-W ENV is enriched in active MS brain lesions [8, 10] and peripheral blood mononuclear cells [11, 12]. A pivotal mechanistic link exists: EBV infection transactivates HERV-W envelope expression, potentially bridging this exogenous viral trigger with endogenous retroviral pathology in MS [13, 14]. Nanoscale extracellular vesicles (EVs), mediating intercellular communication via transfer of proteins, miRNAs, and other cargo [15, 16], are emerging as key vectors in MS pathogenesis [17]. Notably, exosomes can carry pathogenic proteins from exogenous viruses (e.g., EBV) and endogenous retroviruses [18, 19]. Our group recently demonstrated the presence of EBV-encoded proteins on circulating exosomes in MS patients using a validated isolation and flow cytometry pipeline [20]. Building on this foundation, we investigated the presence of pHERV-W on serum exosomes and its clinical relevance in MS.

Materials and methods

Study population

This study was approved by the Institutional Review Board for Research at the American University of Beirut (protocol number IM.SK1.08). Following written

informed consent, venous blood samples were collected from all participants. Serum was isolated and processed for exosome extraction using previously established protocols [20]. The study included three cohorts. Cohort 1 was previously described in [20] and included 12 healthy controls, 14 stable MS patients, and 15 patients with active disease. Active RRMS is characterized by either a clinical relapse or the presence of gadolinium-enhancing (Gd+) lesions on MRI within the previous month. Cohort 2 consisted of patients with relapsing-remitting multiple sclerosis (RRMS), diagnosed according to the revised McDonald criteria [21]. This longitudinal cohort included 15 patients sampled at three timepoints: before disease activity (Pre) (192 ± 89 days), at the time (between 0 and 21 days) of disease activity (Att), defined as either a clinical relapse or the presence of Gd + lesions on magnetic resonance imaging (MRI), and after recovery (Post) (222 ± 108 days). Cohort 3 is a retrospective longitudinal cohort with 25 recently diagnosed, RRMS patients in whom baseline exosomal profiles were linked to clinical outcomes recorded over a follow-up period of three years. Demographic and clinical characteristics of the last two cohorts are detailed in Table 1 (as the characteristics of the first cohort were previously reported [20]).

Immunophenotyping of serum exosomes for HERV and EBV markers

Serum-derived exosomes were isolated and validated according to previously published protocols [20]. A total of 150 micrograms of exosomal proteins were used for surface staining. For the detection of pHERV-W ENV, exosomes were incubated with a biotinylated monoclonal antibody specific to the pHERV-W envelope protein (GENEURO-01-Biot), followed by secondary labeling with Streptavidin conjugated to Alexa Fluor 488 (Thermo Fisher Scientific). To assess EBV latent membrane protein 2 A (LMP2A) expression, exosomes were also stained using a mouse monoclonal anti-LMP2A antibody directly conjugated to allophycocyanin (APC) (Santa Cruz Biotechnology). Following staining, exosomes were thoroughly washed, pelleted by centrifugation, and resuspended in phosphate buffered saline (PBS) for flow cytometric analysis. Samples were analyzed using a BD FACSAria SORP cell sorter at AUB Faculty of Medicine's Flow Cytometry Core Facility, applying previously optimized instrument settings [20] and BD FACSDiva software version 6.0. A minimum of 10,000 events were acquired per sample to ensure reliable quantification. All samples were processed using the same standardized staining protocols and gating strategies, with appropriate controls (e.g., unstained, single-stained samples, stained minus primary, etc.) included in each run to ensure consistency in signal intensity. It is to be noted that while the samples were run on separate or consecutive days,

Table 1 Patients' demographics and clinical characteristics

Characteristic	Cohort 2	Cohort 3
No.	15	25
Female: Male	9: 6	15: 10
Age (yr), mean $\pm$ SD	33 ± 8.4	32 ± 9
Disease duration (yrs), mean $\pm$ SD	6.9 ± 0.8	4.4 ± 0.9
EDSS, mean $\pm$ SD	2.8 ± 1.9	1.5 ± 0.8
DMTs at sampling	Interferon beta therapy (5) S1P receptor modulator (5) Anti-CD20 antibody (2) No Treatment (3)	Interferon beta therapy (11) S1P receptor modulator (4) Other (4) No Treatment (6)

DMTs disease modifying therapies, S1P receptor modulator, Sphingosine-1-Phosphate receptor modulator, SD standard deviation, EDSS Expanded Disability Status Scale

several were replicated across experiments, serving as internal controls. Additionally, the samples were randomized into blocks, each block containing samples from all groups to ensure equal representation across conditions. Time-point samples from each subject were tested in a single experiment. The experiments were conducted with consistent setups and instrument settings, including sample handling, PMT voltage settings, and a standardized gating strategy, all following established protocols to ensure reproducibility. Replicate staining runs of identical samples yielded consistent results, with intra-assay coefficients of variation (CVs) remaining below 5% and inter-assay CV of 7.9%. The exosome verification workflow was comprehensively validated in our previous study [20], where exosomal identity was confirmed by Western blotting, bead-based flow cytometry, electron microscopy, detergent sensitivity testing, and dynamic light scattering. Quality control is done using the Exo-Check™ antibody array and bead-based flow cytometry to verify exosomal characteristics and maintain assay reproducibility. Flow cytometry data were processed and analyzed using R (version 4.3.2) and RStudio (version 2023.09.1 + 494) with the aid of several specialized packages, including flowCore, ggplot2, dplyr, and tidyr. Fluorescence intensity values were log-transformed to normalize skewed distributions and enhance interpretability across samples. Visualization was performed within the R environment to assess marker expression and identify cohort-specific trends.

Multiplex flow cytometric profiling of exosomal surface markers

Serum-derived exosomes from cohort 2 were analyzed using the MACSPlex Exosome Kit (Miltenyi Biotec) that employs 39 polystyrene capture beads labeled with various concentrations of Fluorescein Isothiocyanate (FITC) and Phycoerythrin (PE), enabling the identification of 37 distinct EV surface markers, along with 2 isotype controls, via flow cytometry. Each bead population is individually associated with a specific capture antibody for a respective surface epitope, as detailed in Table 2. After the removal of unbound EVs, those bound to beads were stained with a combination of commonly utilized EV markers, employing APC-coupled detection antibodies

targeting CD9, CD63, and CD81. Semi-quantitative analysis of EV surface markers was then done through the assessment of APC mean fluorescence intensity (MFI) across the diverse bead populations. Briefly, exosomes were subsequently diluted with MACSPlex buffer, resulting in a final volume of 120 µL and a total protein concentration of 10 µg. To ensure the precise removal of unbound and aggregated EVs, the overnight 0.22 µm MACSPlex filter plate protocol was followed according to the manufacturer's instructions. Data analysis was performed utilizing a BD FACSCanto II instrument at KU Leuven Flow and Mass Cytometry Core Facility, with a minimum of 10,000 events recorded per sample. The MACSPlex EV IO setup beads were used for instrument calibration to ensure accurate discrimination among the distinct bead populations. Before each acquisition, PMT voltages were adjusted so that the brightest bead populations reached 10^5 in the PE and FITC channels. As the beads lack fluorescence in the APC channel, the voltage of the APC channel was set to position the intrinsic autofluorescence of the beads at 10^2 , ensuring consistent signal amplification across acquisitions. Single beads were gated to eliminate debris and bead doublets, allowing for the determination of the MFI of the APC signal for each specific bead population. Subsequently, signal values were background-corrected by subtracting the MFI of the control sample (non-EV buffer). Data analysis was conducted using BD FACSDiva software version 6.1.3. To ensure the reliability of signal interpretation, we applied a stringent filtering step: markers detected in fewer than 20% of the samples—meaning they had expression levels below the detection threshold in 80% or more of the patients—were excluded from further analysis. This filtering was applied across all MACSPlex markers before statistical testing.

Flow cytometric profiling of pHERV-W ENV in immune cells and serum-derived exosomes

Peripheral blood mononuclear cells (PBMCs) and serum samples were collected from the same individuals in cohort 2 to assess the expression of pHERV-W ENV at both cellular and exosomal levels. PBMCs were isolated by density gradient centrifugation and stained with a viability dye followed by a panel of fluorochrome-conjugated monoclonal antibodies against CD3, CD8, CD14, and CD19 to define the main immune subsets, along with a biotinylated antibody targeting pHERV-W ENV followed by Streptavidin-Alexa Fluor 488 conjugate (see Supplementary Table 1, Additional file 1). After standard surface staining, cells were washed and resuspended in PBS. Data were acquired using a BD FACSAria SORP cell sorter and analyzed with BD FACSDiva software version 6.0. In parallel, serum-derived exosomes from the same individuals were isolated and stained as previously

Table 2 MACSPlex capture beads coated with antibodies to 37 EV surface markers plus 2 isotype controls

REA Control	mIgG1 Control	SSEA-4	CD133/1	CD209	CD20
CD3	CD105	HLA-ABC	CD29	CD9	CD14
CD4	CD2	CD63	CD69	MCSP	CD24
CD19	CD1c	CD40	CD142	CD146	CD86
CD8	CD25	CD62P	CD45	CD41b	CD44
HLA-DRDPDQ	CD49e	CD11c	CD31	CD42a	CD326
CD56	ROR1	CD81			

described [20]. Dimensionality reduction was performed using t-distributed stochastic neighbor embedding (t-SNE) with standard parameters: perplexity = 30, theta = 0.5, and max_iter = 1000, as implemented in the Rtsne package. After excluding debris and dead cells, all available parameters, including scatter and fluorescence channels, were included in the analysis. The resulting t-SNE plots were used to visualize immune cell clusters and explore pHERV-W ENV expression across populations. Clusters were subsequently identified using k-means, and phenotypes were assigned based on canonical lineage markers (CD14 for monocytes, CD19 for B cells, CD3 for T cells, CD8 for cytotoxic T cells). Correlations between cellular and exosomal pHERV-W ENV expression were evaluated using matched sample comparisons and visualized within R.

Statistical analysis

All statistical analyses were conducted using R (version 4.3.2) and RStudio (version 2023.09.1 + 494). For the cross-sectional analysis, comparisons among healthy controls, stable RRMS patients, and active RRMS patients were performed using one-way ANOVA followed by post hoc multiple comparisons using Tukey's honestly significant difference (HSD) test to identify significant pairwise differences. For cohort 2, paired t-tests were used to compare the expression of pHERV-W ENV and LMP2A across the three timepoints (Pre versus (vs.) Activity, Activity vs. Post, and Pre vs. Post). Protein expression levels were visualized using heatmaps to capture patient-specific and cohort-wide trends. Markers showing paired t-test $p < 0.1$ were retained for further analysis. For these selected variables, delta values ($\Delta = \text{Activity} - \text{Pre}$) were calculated per individual, and Pearson correlation was applied to identify co-regulated changes in exosomal marker expression during the transition from pre-activity to activity. For cohort 3, the subjects were stratified into High and Low pHERV-W ENV expression groups based on the median expression value (cutoff = 1.36). For each clinical outcome, the standardized mean difference (SMD) was computed between groups, and associated p-values were reported to evaluate statistical significance. For all paired comparisons and correlation analyses, we applied Benjamini–Hochberg false discovery rate (FDR) correction separately across the families of tests, including Pre versus Activity, Activity versus Post, Pre versus Post, and delta-correlation analyses. No evidence of disease activity (NEDA) loss, relapse count, EDSS score, and neuro performance measures (Symbol Digit Modalities Test; SDMT, 25-Foot Walk Test; 25FWT, 9-Hole Peg Test; 9HPT) were among the variables assessed. Kaplan–Meier survival curves were generated to visualize time to NEDA loss by biomarker-defined categories. To assess the independent association between

category and NEDA loss while adjusting for potential confounders (age, sex, and treatment), we fitted a multivariable Cox proportional hazards model. The survival time was defined as the duration of follow-up in years, with patients censored at 3 years if NEDA was not lost. NEDA loss was defined as a binary event using the variable 'ever lost NEDA'. We defined evidence of disease activity (EDA) as any clinical relapse or the presence of new lesions (with or without gadolinium enhancement) on MRI. NEDA status was defined as the absence of MS relapse, no new demyelinating lesions on MRI, and no EDSS progression confirmed at 6 months. EDSS progression was defined as an increase of 1.5 points if the baseline EDSS score was 0, 1 point if the baseline EDSS score was between 1 and 5.5, or 0.5 points if the baseline EDSS score was above 5.5. The severity of the relapse was defined as mild (no reported effect on activities of daily living (ADL)), moderate (some interference with ADL), or severe (disabling interference with ADL).

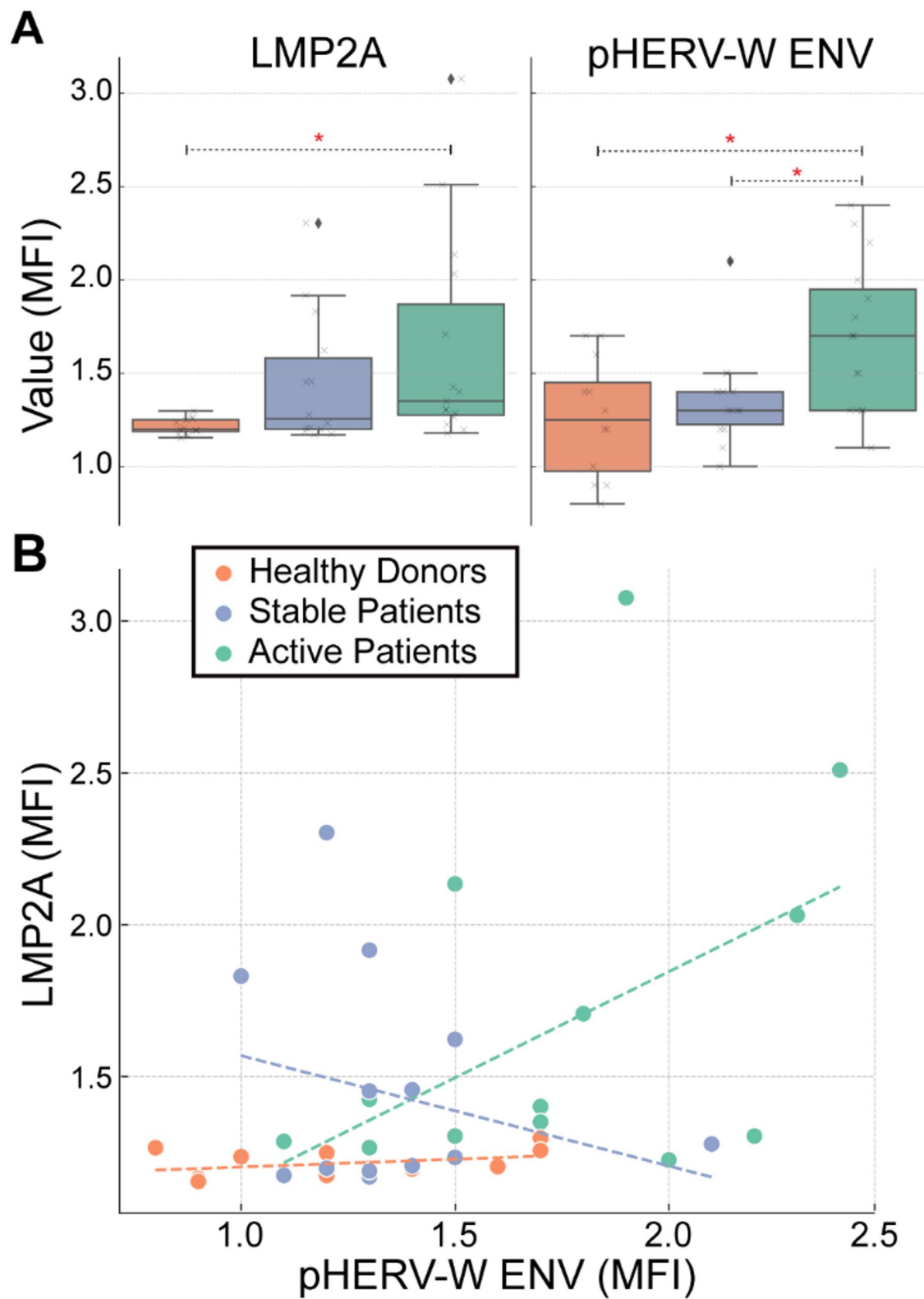
Results

Exosomal expression of pHERV-W ENV and LMP2A in MS patients

Exosomal pHERV-W ENV expression was quantified via flow cytometry in serum samples from healthy donors, stable MS patients, and active MS patients. As shown in Fig. 1A, exosomal pHERV-W ENV levels were significantly elevated in the active MS group compared to healthy donors ($p < 0.05$). Stable MS patients exhibited intermediate levels of expression, slightly higher than those of healthy donors, but the difference did not reach statistical significance ($p = \text{not significant, n.s.}$). LMP2A expression was also measured in the same exosomal fractions. While exosomes from active MS patients expressed higher LMP2A compared to those of healthy controls, the differences were not statistically significant compared to the stable MS group (Fig. 1A). Correlation analysis revealed a significant positive association between pHERV-W ENV and LMP2A expression in active MS patients (Pearson's $r = 0.52$, $p < 0.05$), as illustrated in Fig. 1B. These results demonstrate that pHERV-W ENV expression in serum-derived exosomes is significantly increased in patients with active MS and that it correlates with EBV-LMP2A expression. In contrast, this association is weak or absent in stable patients and healthy individuals (Fig. 1B).

Dynamics of Exosomal pHERV-W ENV and LMP2A expression in relation to disease activity

To investigate temporal changes in viral protein expression with disease activity, serum-derived exosomes were isolated from 15 RRMS patients (Cohort 2). Blood samples were collected at three timepoints: before activity (Pre), during activity, and after activity (Post). Flow

**Fig. 1** (See legend on next page.)

(See figure on previous page.)

Fig. 1 Comparative analysis of exosomal pHERV-W ENV and LMP2A expression across healthy donors and MS patients. **A** Exosomal pHERV-W ENV and LMP2A expression (quantified as MFI) in healthy donors ($n = 12$), stable MS patients ($n = 14$), and active MS patients ($n = 15$). Active patients exhibit elevated exosomal pHERV-W ENV levels compared to both stable patients and healthy controls, and elevated exosomal LMP2A compared to healthy controls. **B** Comparative MFI values for exosomal LMP2A and pHERV-W ENV across the three groups. Horizontal bars or symbols (e.g., dots) represent group-wise data distributions. Statistical significance (e.g., $*p < 0.05$) between active patients and other groups is denoted with asterisks (if applicable), Tukey's post hoc test. Error bars indicate standard deviation or interquartile range

cytometry was used to quantify expression of pHERV-W ENV and LMP2A, reported as MFI. Representative flow cytometry plots from five patients (P16, P17, P18, P24, and P35) are shown in Fig. 2A, highlighting interindividual variation in markers expression. Some patients (e.g., P16, P18) showed a marked increase in pHERV-W ENV expression during disease activity, while others demonstrated only modest changes or none (e.g., P40, P86). LMP2A expression also varied between individuals, with no consistent pattern across all patients. Quantitative longitudinal trends are presented in Fig. 2B. pHERV-W ENV expression exhibited a consistent temporal pattern across the cohort, with levels peaking around the time of disease activity (day 0) ($p < 0.01$) and declining during the post-activity phase ($p < 0.05$). LMP2A expression also increased during disease activity in several patients; however, its expression trajectory was more heterogeneous, with no significant differences detected across timepoints ($p = \text{n.s.}$).

We investigated whether the level of HERV-W ENV expression could predict the timing of disease activity onset. There was a moderate inverse correlation between pHERV-W ENV levels and the number of days until activity onset ($r = -0.49$, $p = 0.06$), suggesting that higher pre-attack expression levels were associated with a shorter time to disease activity. Although this finding did not meet conventional statistical significance, the trend indicates potential predictive relevance. Linear regression modeling estimated a slope of -0.0131 MFI/day, corresponding to an acceleration of disease activity by approximately 76.3 days per unit increase in pHERV-W ENV expression (Fig. 3).

Longitudinal profiling of exosomal surface markers in relation to disease activity

To evaluate phase-dependent changes in exosomal surface marker expression around the time of disease activity, we analyzed a panel of 37 EV markers, including immune regulators (CD40, CD45, HLA-DR/DP/DQ), platelet and endothelial markers (CD42a, CD31, CD62P), and canonical exosome proteins (CD63, CD9, CD81). Markers detected in $\geq 20\%$ of samples were retained for longitudinal comparisons. CD40, an immune costimulatory molecule, and CD42a, a platelet-specific marker, were significantly elevated during disease activity compared to pre-disease activity ($p < 0.05$), suggesting activation of immune and vascular compartments. CD63,

involved in exosome biogenesis and trafficking, showed a trend-level increase ($p < 0.1$). Modest but non-significant elevations were observed in adhesion molecules CD31 and CD62P ($p = 0.07\text{--}0.09$). After Benjamini–Hochberg False Discovery Rate (FDR) correction only pHERV-W ENV remained significant ($q < 0.05$). Thus, CD40 and CD42a should be interpreted as nominal findings, consistent with possible but not statistically robust relapse-associated changes. CD63 exhibited a trend-level increase (raw $p < 0.1$), and modest non-significant elevations were observed for adhesion molecules CD31 and CD62P ($p = 0.07\text{--}0.09$). These results are summarized in Fig. 4. Markers with nominal increases during relapse (e.g., CD40, CD42a, CD63) tended to remain elevated in the post-relapse phase, suggesting sustained immune and vascular activation, though without surviving multiplicity correction. In contrast, multiple immune regulators (CD14, CD45, HLA-ABC) and exosome identity markers (CD9, CD81) remained stable across all phases, indicating that observed changes were selective rather than reflecting broad shifts in exosome abundance or cell-of-origin profiles. Correlation analyses were also adjusted for multiple testing using FDR correction. Expression of pHERV-W ENV remained significantly correlated with CD40 ($r = 0.64$, $q < 0.05$), consistent with coordinated immune–viral upregulation. Hierarchical clustering of the correlation matrix grouped markers by likely cellular origin: platelet-derived (CD42a, CD41b), immune-related (CD40, HLA-DR/DP/DQ), endothelial adhesion molecules (CD31, CD62P), and exosome-associated proteins (CD63, CD133/1), suggesting relapse is accompanied by parallel activation of distinct cellular programs reflected in exosome cargo (see Supplementary Fig. 1, Additional file 2).

Cellular distribution of pHERV-W ENV and its association with exosomal expression

To identify the cellular origin of circulating exosomal pHERV-W ENV, we measured intracellular expression levels of this protein in PBMC subsets from a cohort of 14 RRMS patients (Cohort 2). Flow cytometry was used to quantify pHERV-W ENV (MFI) within four canonical subsets: CD3⁺CD8[−] T cells (primarily CD4⁺ T cells), CD8⁺ cytotoxic T cells, CD14⁺ monocytes, and CD19⁺ B cells. To validate subset identification and ensure precise gating, high-dimensional flow data were projected using t-SNE rather than relying on a traditional hierarchical

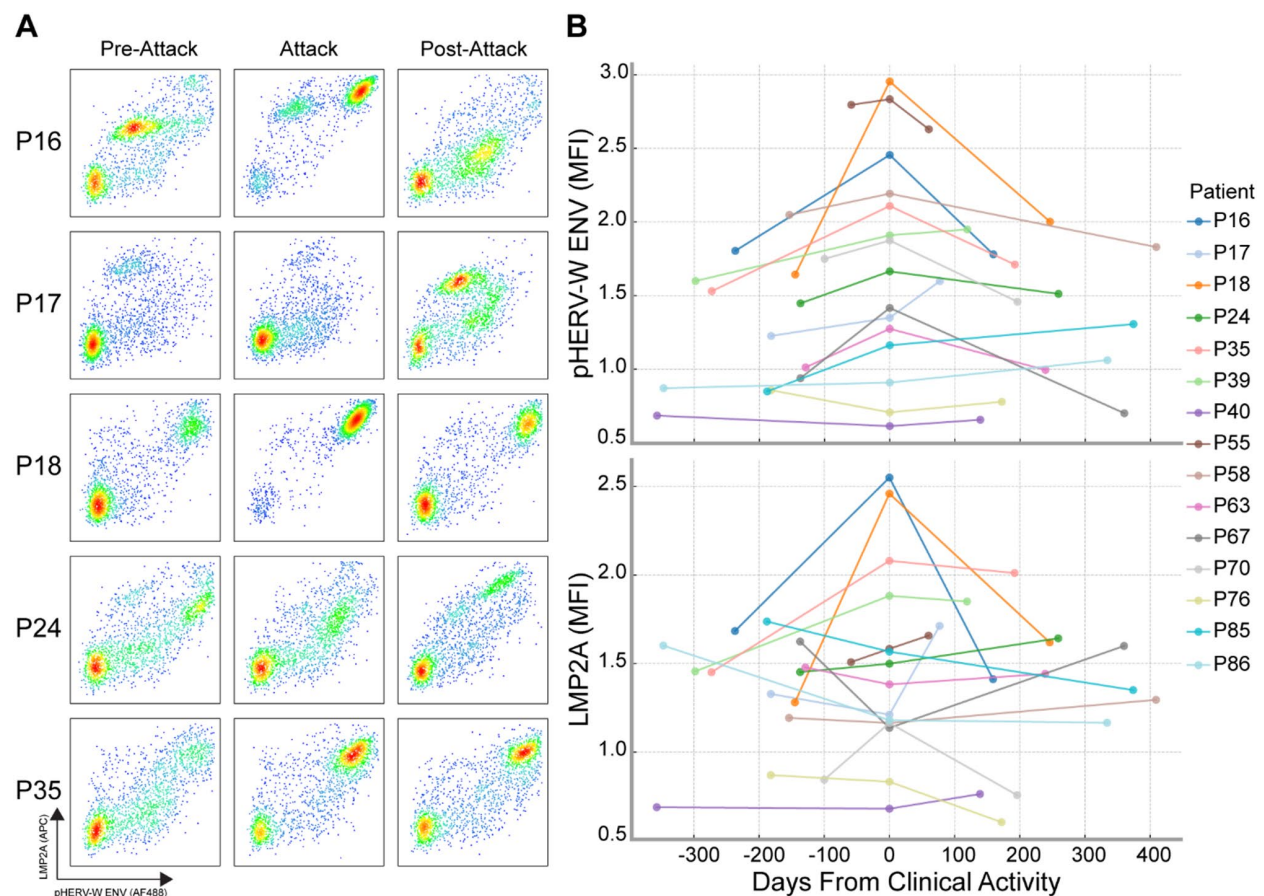


Fig. 2 Longitudinal dynamics of exosomal pHERV-W ENV and LMP2A expression in MS patients. **A** Flow cytometry histograms depicting exosomal detection of pHERV-W ENV and LMP2A MFI in 5 representative patients' samples (P16, P17, P18, P24, P35) collected before, during, and after episodes of disease activity. Fluorescence intensity reflects the marker's abundance, with variability observed across individuals. **B** Longitudinal quantification of pHERV-W ENV (MFI, AF488) and LMP2A (MFI, APC) expression levels in 15 MS patients (P16–P86), aligned temporally relative to the time of disease activity (day 0). Negative timepoints represent pre-activity intervals; positive timepoints denote post-activity phases

gating strategy. As shown in Fig. 5A, pHERV-W ENV was detected in all four populations, with notable differences in expression magnitude and distribution. Monocytes (CD14⁺ cells) exhibited the highest median intracellular expression values across the cohort, followed by CD19⁺ B cells. T lymphocytes (CD3⁺ cells) consistently displayed the lowest MFI values, indicating limited intracellular expression in this subset. Inter-individual variability was observed in the dominant expressing subset: in some subjects, CD19⁺ B cells showed the highest intracellular MFI, while in others, CD14⁺ monocytes were predominant (Fig. 5B). This variability suggests that the cellular source of pHERV-W ENV expression may differ between individuals and could be influenced by immune status, disease phase, or activation state. Importantly, intracellular expression of pHERV-W ENV in CD14⁺ monocytes correlated most strongly with exosomal pHERV-W ENV levels in matched serum samples ($r=0.68$, $p=0.0026$). A significant positive correlation was also observed for

CD19⁺ B cells ($r=0.59$, $p=0.014$). In contrast, no significant correlations were found for CD3⁺CD8⁺ T cells ($r=-0.29$, $p=0.25$) or CD8⁺ T cells ($r=-0.08$, $p=0.75$) (Fig. 5B). Given their enrichment for pHERV-W ENV, monocytes and B cells represent likely contributors to the circulating exosomal pool in this cohort, though definitive cellular origins remain to be confirmed.

Predictive and prognostic value of exosomal pHERV-W ENV expression in relapsing-remitting multiple sclerosis

To assess the long-term prognostic value of baseline pHERV-W ENV expression in serum-derived exosomes, we investigated a separate longitudinal cohort of 25 RRMS patients (cohort 3) followed for a median of 36 months. Patients were stratified into high ($n=12$) and low ($n=13$) expression groups based on exosomal pHERV-W ENV median expression value (cutoff=1.36). Demographic characteristics, including age, sex distribution, and treatment assignment, were comparable between

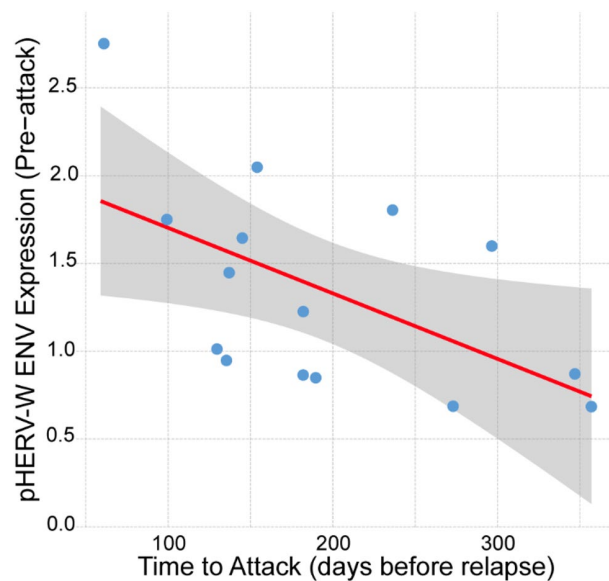


Fig. 3 Exosomal pHERV-W ENV in relation to the time of onset of disease activity. Individual data points (blue circles) represent pHERV-W ENV levels plotted against time. The red line is the mean, and the shaded gray region represents the confidence intervals. Correlation ($r = -0.488$, $p = 0.06$, $n = 15$), and regression coefficient = 76.3 days per MFI unit

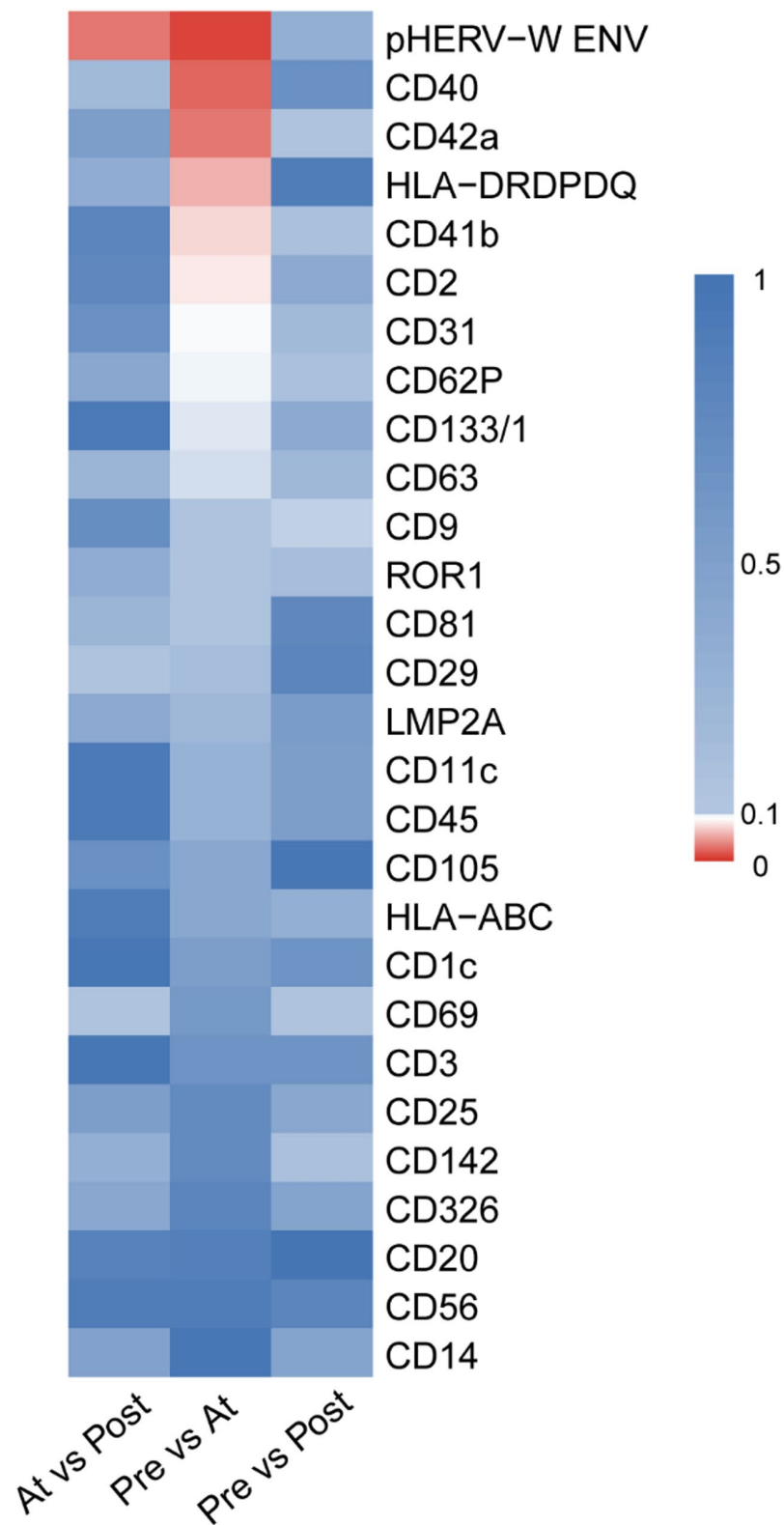
groups (Table 3). To assess the independent association between biomarker-defined patient groups and time to loss of NEDA, we performed a multivariable Cox proportional hazards regression. The primary predictor was the pHERV-W ENV category variable, reflecting biomarker grouping. The model was adjusted for potential confounders such as age, sex, and treatment group (initial and last). The treatments were divided into three categories based on efficacy: high, moderate, and low (see Supplementary Table 2, Additional file 3). Disease-modifying therapies (DMTs) along with the corresponding number of patients are presented for each category at both the initial visit (when sampling occurred) and the last visit (during the follow-up after three years) in Supplementary Table 2, Additional file 3. Results showed that the biomarker-defined category was significantly associated with a higher hazard of NEDA loss (adjusted Hazard Ratio = 8.29 [95% CI: 1.30, 52.73], $p = 0.025$), indicating that patients in the High pHERV-W ENV category were more than eight times as likely to lose NEDA status over time, independent of age, sex, and treatment (Fig. 6A).

Proportional hazards assumptions were tested with Schoenfeld residuals and were not violated (global $p = 0.53$). Discrimination of the model was good (apparent C-statistic = 0.81, optimism-corrected C-statistic ≈ 0.70). Leave-one-out influence analyses preserved the pHERV-W ENV effect, with hazard ratios ranging between 6.37 and 28.93. Calibration at 3 years showed close agreement between predicted and observed risks across tertiles (low: 0.14 predicted vs. 0.22 observed; mid:

0.71 vs. 0.75; high: 0.95 vs. 1.00). A minimalist model including pHERV-W ENV, age, and sex yielded consistent results (HR = 5.08, 95% CI 1.62–15.94, $p = 0.005$). SMD analysis identified total disease activity events as the most discriminant variable ($p < 0.01$), while measures of physical disability and cognitive performance, including EDSS, 25FWT, 9HPT, and SDMT, did not significantly differ between groups (Fig. 6B).

Discussion

Building upon our foundational work demonstrating the presence of EBV markers on exosomes in MS patients [20], the current study provides compelling evidence that exosomal pHERV-W ENV expression is significantly increased during active MS and serves as a dynamic, clinically relevant biomarker of disease activity. These findings reinforce the mounting evidence implicating endogenous retroviral elements as potential drivers of inflammation in MS. For the detection of pHERV-W ENV, exosomes were incubated with a biotinylated monoclonal antibody specific to the Human endogenous Retrovirus-W Envelope protein (GENEURO-01-Biot, GeNeuro) with demonstrated specificity [10]. Clinically, it has been reported that the humanized version of this antibody (GNbAC1/temelimab) binds the extracellular ENV domain with high affinity and has been safely administered in clinical trials in MS patients [22, 23]. It has also been evaluated in clinical trials for type 1 diabetes, where it was similarly demonstrated to be safe and well-tolerated [24]. Our initial finding of increased exosomal pHERV-W ENV expression levels at the time of relapse or evidence of GD + lesions on MRI aligns with recent literature reporting upregulation of pHERV-W ENV expression in MS patients, particularly in active disease states [19, 25]. The positive correlation between pHERV-W ENV and LMP2A expression specifically in active MS patients (Pearson's $r = 0.52$, $p < 0.05$) hints at a synergistic viral interaction. This is relevant given our prior work showing EBV markers on exosomes, and EBV's well-known capacity to transactivate HERV elements [26]. This correlation is consistent with the emerging concept that exogenous viral infections such as EBV, may act as critical triggers for the de-repression and activation of endogenous retroviruses, thereby igniting and fueling the inflammatory cascade [27, 28]. Mechanistically, pHERV-W ENV is a well-established driver of neuroinflammation and plays a role in remyelination failure, primarily through its agonistic interaction with Toll-like receptor 4 (TLR4) on glial cells, leading to pro-inflammatory cytokine release and direct axonal damage [8, 29]. The detection of these viral proteins within exosomes as shown here, suggests a sophisticated and efficient mechanism for their

**Fig. 4** (See legend on next page.)

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Fig. 4 Exosomal marker expression. Heatmap of paired t-test results comparing exosomal marker expression at the time of disease activity vs. after (column 1), before disease activity vs. during (column 2), and before vs. after disease activity (column 3) in RRMS patients ($n=15$). Markers retained after stringent filtering (detected in $\geq 20\%$ of samples) include immune regulators (CD40, CD45, HLA-ABC), adhesion molecules (CD31, CD29), canonical exosome markers (CD63, CD9, CD81), and viral proteins (pHERV-W ENV, LMP2A). Color intensity represents raw p-values using a non-linear blue–white–red gradient, with red indicating stronger significance ($p < 0.05$). Among these, pHERV-W ENV, CD40, and CD42a reached nominal significance ($p < 0.05$); however, only pHERV-W ENV remained significant after FDR correction. Clinical take-home: relapse is marked by a viral (pHERV-W ENV) signal with concurrent but less robust immune (CD40) and platelet (CD42a) activation, some of which persists post-relapse

dissemination and impact across the blood-brain barrier, effectively contributing to lesion development and the clinical manifestations of active disease [30]. Collectively, these findings position exosomal pHERV-W ENV as a potential biomarker of disease activity but also as a mechanistic contributor to MS pathogenesis. While our data support HERV-W ENV as a biomarker candidate in MS, it is unclear if this dynamic is exclusive to HERV-W or if other HERV families may also be involved. Multiple HERV families may contribute to neuroinflammatory and neurodegenerative diseases, albeit with varying disease specificity and clinical relevance [4, 5]. For instance, human endogenous retrovirus K (HERV-K) has been implicated in several neurological disorders, most notably amyotrophic lateral sclerosis (ALS), as well as Alzheimer's disease and schizophrenia, indicating its broader involvement in neurodegenerative processes [31, 32]. In the context of MS, HERV-K has been detected at modestly increased levels in certain tissues; however, these changes are subtle and not statistically or clinically robust [33]. Furthermore, the presence of a well-characterized polymorphic variant, HERV-K113, has shown no significant association with MS susceptibility [34]. Therefore, overall evidence supporting HERV-K as a reliable biomarker in MS remains weak and inconclusive [33, 34]. Similarly, while HERV-H/F has been reported as elevated in PBMCs of patients with active MS [35, 36], strong or consistent evidence associating it with the disease remains limited. In contrast, HERV-W demonstrates far stronger and more consistent associations with MS pathophysiology than other endogenous retroviruses [4, 31, 37] and has progressed to targeted clinical trials [23]. Thus, it is recognized as a potential biomarker for monitoring therapeutic response in the context of treatments targeting HERV-W activity [31]. In Phase II clinical trials, neutralization of pHERV-W ENV with temelimab was associated with reductions in MRI-detected brain lesions, suggesting potential neuroprotective and anti-neurodegenerative effects [23]. The consistent and significant peak in pHERV-W ENV expression at the time of disease activity illustrates its tight association with acute neuroinflammatory events. This transient upregulation mirrors previous reports connecting elevated pHERV-W ENV levels with active neuroinflammation during MS flares [38]. Interestingly, while pHERV-W ENV

displayed a coherent temporal trajectory, LMP2A levels fluctuated more heterogeneously, suggesting that while EBV might influence HERV activation, the exosomal expression of LMP2A may be more individualistic, potentially modulated by the nuances of latent EBV reactivation kinetics or host-specific immune variables [39]. This distinction further underscores the utility of pHERV-W ENV as a particularly sensitive and activity-specific marker in RRMS. Nonetheless, our findings should be interpreted as exploratory and hypothesis-generating, as causality cannot be inferred. While our assay demonstrated acceptable repeatability and reproducibility, this study represents a proof-of-concept with preliminary validation. A full analytical validation package will be essential in future prospective multicenter studies before translation into clinical use.

We demonstrated a significant increase in exosomal CD40 expression at the time of disease activity, aligning with the well-established role of CD40-CD40L signaling in MS lesion development and neuroinflammation [40, 41]. CD40 expression showed a downward trend post-activity, but did not return to baseline. This contrasts with the sharper, more transient dynamics observed for pHERV-W ENV expression. Concurrently, we observed a significant elevation in the platelet-specific marker CD42a, along with trends in CD41b and CD62P, indicating active platelet involvement during disease activity, which may contribute to blood-brain barrier disruption and vascular inflammation, a growing area of focus in MS pathogenesis [42, 43]. The observed positive correlation between exosomal pHERV-W ENV and CD40 suggests a synchronized upregulation of retroviral and immune signaling axes, both of which are known to converge on pro-inflammatory pathways implicated in MS flare-ups [8, 10]. This exosomal signature is consistent with previous work demonstrating increased surface expression of pHERV-W ENV and activation markers on circulating immune cells during MS activity [35, 44–46]. In conclusion, these data support the model of MS disease activity as a temporally layered and cell-type-specific response, in which exosomal pHERV-W ENV acts as a transient viral instigator, while immune and vascular elements contribute to and sustain the broader inflammatory milieu.

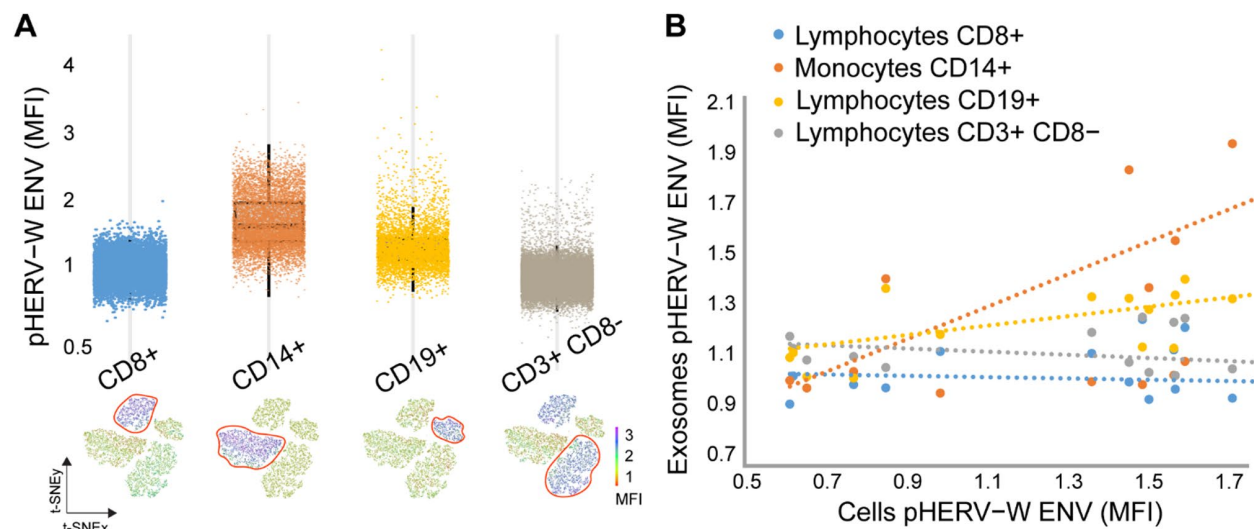


Fig. 5 Cellular expression patterns of pHERV-W ENV in peripheral immune subsets and correlation with exosomal levels. **A** Boxplot of pHERV-W ENV expression across immune subsets in RRMS PBMCs ($n=14$). Cells are clustered by lineage markers (CD3, CD8, CD14, CD19) and presented as t-SNE plots with color intensity representing median fluorescence intensity (MFI) of each. Monocytes (CD14⁺) show the highest overall median expression, followed by B cells (CD19⁺). **B** Correlation between cellular pHERV-W ENV MFI in immune subsets and matched serum exosomal pHERV-W ENV levels. Pearson correlation analysis showed significant positive correlations for monocytes (CD14⁺, $r=0.68$, $p=0.0026$) and B cells (CD19⁺, $r=0.59$, $p=0.014$). No significant correlations were observed for T lymphocyte subsets (CD8⁺, $r=-0.08$, $p=0.75$; CD3⁺ CD8⁻, $r=-0.29$, $p=0.25$)

Table 3 pHERV-W ENV patient group comparison

Variable	Low pHERV-W ENV (≤ 1.36)	High pHERV-W ENV (> 1.36)	p-value
No.	13	12	
Sex	F: 61.5%; M: 38.5%	F: 58.3%; M: 41.7%	0.87
Age	32.5 $\pm$ 9.25	32.8 $\pm$ 9.30	0.93
Ever lost NEDA	38.5%	91.7%	0.01
Lost NEDA time (yrs)	2.3 $\pm$ 1.1	1.2 $\pm$ 0.9	0.01
Disease activity events	1.6 $\pm$ 0.7	2.4 $\pm$ 1.0	0.03
First EDSS score	1.4 $\pm$ 0.8	1.8 $\pm$ 1.0	0.27
Last EDSS score	1.6 $\pm$ 0.9	2.0 $\pm$ 1.1	0.31
EDSS score change	0.3 $\pm$ 0.9	0.3 $\pm$ 0.9	0.95
25FWT	4.4 $\pm$ 1.4	4.1 $\pm$ 0.9	0.61
SDMT Num	51.8 $\pm$ 12.0	52.5 $\pm$ 18.5	0.91
SDMT Denom	52.7 $\pm$ 11.8	52.7 $\pm$ 18.4	1.00
Dom 9HPT	20.3 $\pm$ 7.5	21.2 $\pm$ 7.1	0.77
Non-Dom 9HPT	21.2 $\pm$ 11.6	25.5 $\pm$ 14.7	0.42
Initial treatment	No treatment: 30.8% Low efficacy: 38.5% Moderate efficacy: 23.1% High efficacy: 7.7%	No treatment: 16.7% Low efficacy: 41.7% Moderate efficacy: 33.3% High efficacy: 8.3%	0.86
Last treatment	Low efficacy: 15.4% Moderate efficacy: 38.5% High efficacy: 46.2%	Low efficacy: 8.3% Moderate efficacy: 25.0% High efficacy: 66.7%	0.58

Analysis shows no baseline differences in age, sex, EDSS, and other clinical variables

Our data suggest that monocytes and B cells are the principal cellular contributors of the circulating exosomal pHERV-W ENV pool in MS patients. This observation resonates with the accumulating evidence that activated B cells, often latently infected with EBV, serve as critical functional reservoirs for HERV-W-related immunopathology, with increased pHERV-W ENV expression reported on their surface during MS flares [35] as well as in untreated MS patients compared to those on DMTs [13]. The process of retroviral protein packaging into exosomes, particularly when originating from potent antigen-presenting cells such as B cells, serves as an efficient pathway for systemic dissemination and distant cellular impact, including passage across the blood-brain barrier [47]. Interestingly, we found that CD14⁺ monocytes express pHERV-W ENV, in contrast with previous transcriptomic studies reporting weaker HERV-W signatures in monocyte-derived populations compared to lymphocytes [46, 48]. One possible explanation for this discrepancy is that transcriptomic approaches measure mRNA levels, which do not always reflect actual protein expression or surface presentation. Our findings support monocytes and B cells as primary cellular origins for circulating exosomal pHERV-W ENV in MS, but further mechanistic studies are required. These findings are pertinent given the transformative success of B-cell depleting therapies in MS, suggesting that B cells' pathogenic role extends far beyond antibody production to encompass crucial immunomodulatory functions, including the

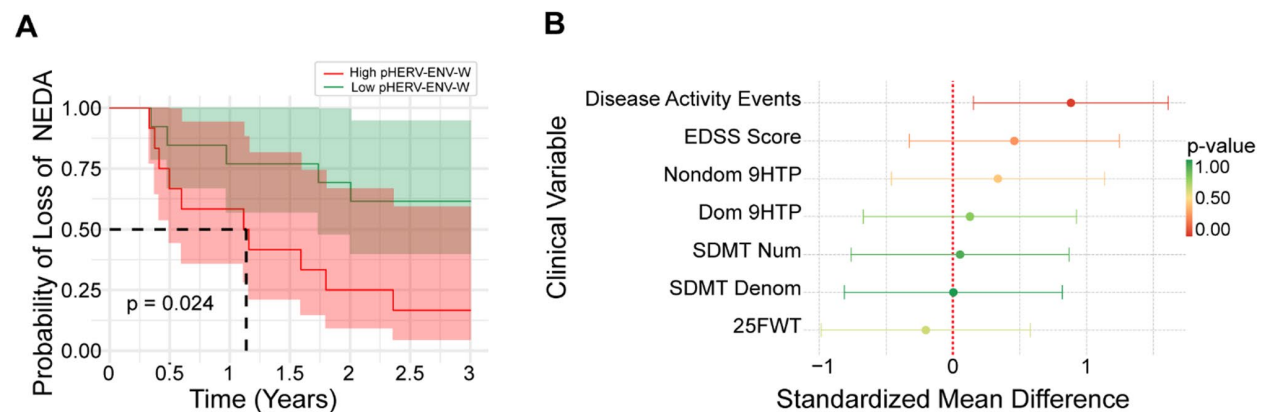


Fig. 6 Prognostic value of exosomal pHERV-W ENV expression in RRMS: Time to NEDA loss. Association between baseline exosomal pHERV-W ENV expression and clinical outcomes in RRMS (Cohort 3, $n = 25$). **A** Kaplan-Meier curve showing time to loss of NEDA status in patients stratified by high vs. low pHERV-W ENV expression. High-expression patients (red) showed significantly earlier NEDA loss compared to low-expression patients (blue) (log-rank $p = 0.024$). **B** SMDs for clinical variables between high- and low-expression groups. Total relapse count during follow-up (disease activity events) was the only variable showing significant discrimination ($p < 0.01$). Physical disability measures (EDSS, 9HPT, 25FWT) and cognitive performance (SDMT) did not differ significantly between groups

potential to propagate neuroinflammation via exosomal pHERV-W ENV.

Finally, our data provide evidence for the utility of exosomal pHERV-W ENV expression as a short-term predictor of disease activity and a long-term prognostic marker in RRMS. High baseline exosomal pHERV-W ENV expression predicted a significantly greater number of disease activity events (mean = 2.42 vs. 1.62; $p < 0.05$) and loss of NEDA status (91.7% vs. 38.5%; $p < 0.05$). Infections and vaccinations could potentially affect pHERV-W ENV expression, and this is a limitation in this study. Additionally, EBV serology and viral load data were not available in this cohort, and this would need to be addressed in future studies.

Neurofilament light chain (NfL) is a widely recognized marker of axonal damage in MS [49, 50]. While serum NfL (sNfL) levels are elevated during MS relapses and correlate with future disease activity and NEDA status [51, 52], pHERV-W ENV also offers a potentially unique advantage by targeting an upstream driver of pathogenesis rather than merely reflecting downstream neuronal injury. Therefore, we propose that pHERV-W ENV complements rather than substitutes sNfL. Future prospective validation studies should combine pHERV-W ENV expression and sNfL measurements in larger cohorts. Given that pHERV-W ENV expression is thought to be inducible by exogenous herpesviruses, such as EBV and Human Herpesvirus 6 A [19], its measurement could serve as an integrated marker reflecting both environmental risk factors and the host's immune status.

In conclusion, our findings suggest that exosomal pHERV-W ENV may serve as a promising biomarker for disease activity. Future research that combines exosomal pHERV-W ENV with other important biomarkers,

such as NfL, could create a more comprehensive and multi-dimensional risk model. This model would be both biologically relevant and clinically actionable for personalized management of MS.

Abbreviations

25FWT	25-Foot Walk Test
9HPT	9-Hole Peg Test
ADL	Activities of Daily Living
APC	Allophycocyanin
CNS	Central Nervous System
DMTs	Disease-Modifying Therapies
EBV	Epstein-Barr Virus
EDA	Evidence of Disease Activity
EDSS	Expanded Disability Status Scale
EVs	Extracellular Vesicles
FITC	Fluorescein Isothiocyanate
Gd+	Gadolinium-enhancing
HERV-W	Human Endogenous Retrovirus Type W
LMP2A	Latent Membrane Protein 2 A
MFI	Mean Fluorescence Intensity
MS	Multiple Sclerosis
NEDA	No Evidence of Disease Activity
NfL	Neurofilament Light Chain
PBMCs	Peripheral Blood Mononuclear Cells
PBS	Phosphate Buffered Saline
PE	Phycoerythrin
pHERV-W ENV	Envelope Protein of Human Endogenous Retrovirus Type W
RRMS	Relapsing-Remitting Multiple Sclerosis
SDMT	Symbol Digit Modalities Test
S1P receptor modulator	Sphingosine-1-Phosphate Receptor Modulator
SMD	Standardized Mean Difference
TLR4	Toll-Like Receptor 4

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

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Author contributions

*The authors (ESS and MFM) contributed equally to this work. E.S.S. and M.F.M. contributed equally to this work, including study conceptualization, data analysis and interpretation, drafting the text and figures, and revising the manuscript for intellectual content. M.F.M. was responsible for study design and data acquisition, and E.S.S. was responsible for statistical analysis and data reconfiguration. L.N. was responsible for data acquisition and analysis, as well as manuscript revision. V.D. performed multiplex profiling experiments and related analyses, and revised the manuscript. J.A. was responsible for clinical data curation and manuscript revision. S.J.K. conceptualized the study, acquired funding, analyzed and interpreted data, and revised the manuscript for intellectual content. All authors have read and approved the final manuscript for submission.

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Data availability

All relevant data are provided in the manuscript and supplementary files. The datasets generated and analyzed during the current study, including raw flow cytometry files (FCS) and processed summary tables, are available from the corresponding author upon reasonable request. Due to patient privacy regulations, data cannot be publicly posted, but will be shared under a data use agreement with qualified investigators.

Declarations

Ethics approval and consent to participate

This study was approved by the Institutional Review Board for Research at the American University of Beirut (protocol number IM.SK1.08), and all participants provided written informed consent.

Competing interests

The authors declare no competing interests.

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